



## ANALYSTLAB AFRICA

Experience Lab Internship Programme

Official Weekly Assignment

**Week 5: HealthConnect Solution Development & Implementation**

### Document Control

|                     |                                                                                   |
|---------------------|-----------------------------------------------------------------------------------|
| Programme           | AnalystLab Africa Experience Lab                                                  |
| Week                | Week 5                                                                            |
| Project             | HealthConnect Clinic Experience Lab                                               |
| Project Title       | Improving Patient Appointment Attendance and Healthcare Support Using Data and AI |
| Version             | 1.0                                                                               |
| Issue Date          | _____                                                                             |
| Submission Deadline | End of Week 5 (Sunday, 11:59 PM WAT)                                              |
| Submission Method   | Official AnalystLab Africa Week 5 Submission Form                                 |

## 1. BUSINESS SCENARIO

HealthConnect Clinic is a fictional healthcare provider that manages appointment-based services.

The clinic is seeking to reduce missed appointments, better understand appointment attendance patterns, improve the use of appointment slots, and provide more effective administrative support to patients.

In Week 4, interns established their track-specific understanding of the HealthConnect problem, reviewed the available project resources, identified their responsibilities, and developed an initial approach.

**Week 5 moves the project from planning into practical implementation.**

Interns are now expected to use the work completed in Week 4 as the foundation for producing tangible outputs for the HealthConnect project.

The overall project question remains:

**How can HealthConnect Clinic use data and AI to reduce missed appointments and improve the patient support experience?**

## 2. PROJECT CONTINUITY

From Week 4 onwards, all interns contribute to the **same HealthConnect Experience Lab project**, while working from the perspective of their respective professional tracks.

Week 4 focused on:

**Problem Understanding → Resource Review → Solution Planning**

Week 5 focuses on:

**Analysis → Development → Initial Implementation**

The work completed in Week 5 should therefore build directly on the intern's Week 4 submission.

Interns should **not restart the project or select a new dataset/project**.

The HealthConnect resources remain the foundation of the project:

- HealthConnect Appointment Dataset
- HealthConnect Data Dictionary
- HealthConnect Clinic Knowledge Base, where applicable

The original project resources must not be overwritten. Any processed, transformed, or derived files should be saved separately and documented.

## 3. WEEK 4 → WEEK 5 TRANSITION

At the end of Week 4, each intern was expected to establish a clear foundation for their work.

Depending on the track, this included:

- defining the problem;
- identifying relevant resources;
- identifying important data or project requirements;
- proposing an approach;
- identifying assumptions, limitations, risks or dependencies; and
- identifying the proposed focus for Week 5.

**In Week 5, interns must now move from “what should we do?” to “let's begin doing it.”**

This means Week 5 should produce **actual analysis, development, designs, prototypes, project artefacts or implementation outputs**, depending on the track.

## 4. ASSIGNMENT OBJECTIVE

The objective of Week 5 is to begin the practical development of the HealthConnect solution by applying the approach established during Week 4.

By the end of Week 5, interns should be able to:

- apply their Week 4 project understanding;
- work with the relevant HealthConnect project resources;
- produce meaningful track-specific outputs;
- document their implementation process;
- identify challenges encountered during development;
- evaluate initial results where applicable;
- collaborate with other project tracks where appropriate; and
- establish a clear foundation for Week 6.

## 5. LEARNING OBJECTIVES

By the end of this assignment, interns should be able to:

- Translate a defined business problem into practical project work.
- Apply appropriate tools and techniques to their professional track.
- Analyse, develop, design or implement components of a real-world project.
- Document technical and analytical decisions.
- Evaluate initial outputs.
- Identify limitations and areas for improvement.
- Communicate progress professionally.
- Work as part of a multidisciplinary project team.

## 6. PROJECT RESOURCES

The following HealthConnect resources are available:

### 6.1 HealthConnect Appointment Dataset

**File:** HealthConnect\_Appointment\_Data.csv

The dataset contains fictional and anonymized appointment records, including information relating to:

- patient demographics;
- appointment details;
- booking information;
- previous appointment history;
- previous no-shows;
- reminder information;
- distance to the clinic;
- waiting time; and
- appointment outcomes.

## 6.2 HealthConnect Data Dictionary

**File:** HealthConnect\_Data\_Dictionary.xlsx

Use this resource to understand the variables contained in the appointment dataset.

## 6.3 HealthConnect Clinic Knowledge Base

**File:** HealthConnect\_Clinic\_Knowledge\_Base.docx

This resource contains approved fictional information about the clinic, including appointment procedures, rescheduling, cancellations, FAQs, safety rules and escalation guidelines.

**Important:** Generative AI solutions must remain within the approved Knowledge Base. Interns must not invent clinic or medical information.

# 7. TOOLS AND TECHNOLOGIES

Interns should use tools appropriate to their assigned track and Week 5 approach.

### Project Management

- Microsoft Excel / Google Sheets
- Microsoft Word / Google Docs
- PowerPoint / Google Slides
- Trello
- Notion
- Jira or similar tools

### Data Analytics

- SQL
- Python
- Power BI
- Tableau

### Data Science

- Python
- Jupyter Notebook / Google Colab
- VS Code
- Pandas
- NumPy
- Matplotlib
- Scikit-learn

### Machine Learning Engineering

- Python

- Jupyter Notebook / VS Code
- Git
- GitHub
- Appropriate ML libraries

### Generative AI

- Python
- Jupyter Notebook / VS Code
- Git / GitHub
- Appropriate LLM tools, APIs or frameworks

These tools are examples rather than a requirement to use every listed technology. This follows the Week 4 approach.

## 8. ASSIGNMENT TASKS

### PART 1: REVIEW YOUR WEEK 4 FOUNDATION

Before beginning the Week 5 practical work, review your Week 4 submission.

Briefly identify:

1. The problem you defined.
2. Your Week 4 proposed approach.
3. The HealthConnect resources relevant to your work.
4. The key assumptions or limitations identified.
5. What you proposed to accomplish in Week 5.

This should be a **brief transition section**, not a repetition of the entire Week 4 submission.

### PART 2: TRACK-SPECIFIC ASSIGNMENT

#### A. PROJECT MANAGEMENT TRACK

##### Week 5: Project Execution & Progress Management

In Week 4, you established the initial project structure, scope, stakeholders, timeline, risks and communication approach.

In Week 5, your responsibility is to move from **project initiation to project execution and monitoring**.

#### Tasks

1. Review the Week 4 project plan.
2. Convert major Week 4 activities into actionable tasks.
3. Assign responsibilities across HealthConnect workstreams.

4. Establish task priorities.
5. Track project progress.
6. Update the project timeline.
7. Monitor identified risks and dependencies.
8. Identify newly emerging project risks.
9. Document project status.
10. Develop an approach for coordinating the different technical tracks.

## Week 5 Output

Prepare a **Project Execution & Progress Package** containing:

- Updated project plan.
- Task tracker.
- Updated timeline.
- Progress/status report.
- Updated risk and dependency register.
- Track coordination plan.
- Summary of major project issues encountered.

## B. DATA ANALYTICS TRACK

### Week 5: Exploratory Analysis, KPI Development & Business Insights

In Week 4, the Data Analytics track reviewed the appointment data, assessed data quality, identified business questions and proposed potential KPIs.

Week 5 now moves into **actual analysis**.

### Tasks

#### 1. Data Preparation

Use the HealthConnect Appointment Dataset and Data Dictionary to:

- inspect the relevant variables;
- check data types;
- identify missing values;
- identify duplicates;
- identify inconsistent values;
- document relevant data-quality observations.

Do not overwrite the original dataset.

#### 2. Exploratory Data Analysis

Investigate appointment attendance and no-show patterns.

Analyse relevant relationships such as:

- appointment characteristics;
- patient characteristics;
- previous appointment history;
- reminders;
- waiting time;
- distance to clinic;
- cancellations;
- no-show outcomes.

### 3. KPI Development

Select **3–5 KPIs** from the potential KPIs identified in Week 4.

For each KPI:

- define it;
- explain why it matters;
- calculate it;
- identify the relevant business question;
- interpret the result.

### 4. Dashboard or Analytical Visualisation

Create an initial analytical dashboard or visual report containing relevant findings.

### 5. Business Insights

Produce at least **5 meaningful insights** and explain their potential implications for HealthConnect.

### Week 5 Output

Submit an **Initial HealthConnect Analytics Report or Notebook** containing:

- Data preparation summary.
- EDA.
- KPI calculations.
- Visualisations/dashboard.
- Key findings.
- Business recommendations.
- Limitations.

## C. DATA SCIENCE TRACK

### Week 5: Data Preparation, Feature Engineering & Baseline Model Development

In Week 4, Data Science interns defined the proposed machine-learning problem, target variable, potential features and initial modelling approach.

Week 5 moves into the first practical modelling stage.

## Tasks

### 1. Review the Week 4 ML Problem Definition

Confirm:

- the prediction problem;
- proposed target;
- candidate features;
- treatment of cancellations;
- modelling objective.

If Week 5 analysis reveals that a decision needs to change, **document and justify the change** rather than silently changing it.

### 2.a Data Preparation

**Prepare the dataset for modelling.**

Address where applicable:

- Missing values.
- Duplicates.
- Incorrect data types.
- Categorical variables.
- Irrelevant variables.
- Inconsistent records.
- Potential data leakage.

### 2.b Data Exploration & Visual Evidence

Use appropriate visualizations to support your data-preparation, feature-engineering, feature-selection and modelling decisions.

Depending on the dataset and analytical need, these may include:

- Histograms or distribution plots.
- Box plots for potential outliers.
- Count plots or bar charts for categorical variables and target distribution.
- Correlation heatmaps.
- Feature–target relationship plots.
- Other appropriate visualizations.

For each **important visualization**, provide a brief interpretation explaining:

- What the visualization shows.
- What you observed.
- How the observation informed a preprocessing, feature engineering, feature selection or modelling decision.

**Important:** Do not create visualizations simply to meet a numerical requirement. Focus on producing **meaningful, decision-supporting visual evidence**.

### 3. Feature Engineering

Create meaningful features that could help predict appointment attendance/no-shows.

For each engineered feature:

- explain what was created;
- explain why it was created;
- explain its expected modelling value.

### 4. Train/Test Strategy

Define an appropriate strategy for separating data into training and testing sets.

Explain why the chosen approach is appropriate for the problem.

### 5. Baseline Model

Develop at least **one baseline classification model** appropriate to the defined target.

Document:

- model selected;
- reason for selection;
- training process;
- prediction results.

### 6. Initial Evaluation

Evaluate the baseline model using appropriate classification metrics.

Where applicable, consider:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix
- ROC-AUC

The focus at this stage is **establishing a reliable baseline**, not achieving the final best-performing model.

### Week 5 Output

Prepare a **Data Science Baseline Modelling Notebook/Report** containing:

- Week 4 problem review.
- Data preparation.
- Feature engineering.
- Train/test strategy.
- Baseline model.
- Initial evaluation.
- Interpretation.
- Modelling limitations.
- Recommendations for Week 6.

## D. MACHINE LEARNING ENGINEERING TRACK

### Week 5: ML Pipeline Development & Initial Implementation

In Week 4, the ML Engineering track defined the expected system inputs and outputs, workflow, architecture, dependencies and implementation approach.

Week 5 begins the implementation of that design.

#### Tasks

1. Review the Week 4 system architecture.
2. Establish the project repository structure.
3. Implement the initial data-processing workflow.
4. Create reusable preprocessing components where appropriate.
5. Implement the initial model workflow or model integration structure.
6. Establish configuration and dependency requirements.
7. Document the pipeline.
8. Implement basic error handling where appropriate.
9. Test the individual pipeline components.
10. Document issues and dependencies.

#### Week 5 Output

Prepare an **Initial ML Pipeline Implementation Package** containing:

- Updated repository.
- Data-processing pipeline.
- Initial model integration/workflow.
- Project structure.
- Requirements/dependencies.
- Pipeline documentation.
- Basic testing evidence.
- Implementation issues and next steps.

The focus is **not deployment** at this stage.

## E. GENERATIVE AI TRACK

### Week 5: Prompt Engineering, Knowledge Integration & Assistant Prototype

In Week 4, the Generative AI track defined the intended users, use cases, supported requests, out-of-scope requests, safety boundaries and escalation approach.

Week 5 moves from design into **initial prototyping**.

#### Tasks

##### 1. Review the Week 4 Assistant Design

Confirm:

- target users;
- supported use cases;
- out-of-scope requests;
- safety boundaries;
- escalation requirements.

##### 2. Prompt Development

Develop prompts for the approved use cases.

Test different prompt structures and document:

- prompt objective;
- prompt used;
- output generated;
- strengths;
- weaknesses;
- improvements made.

##### 3. Knowledge Base Integration

Use the approved HealthConnect Clinic Knowledge Base as the information source for the assistant.

The assistant should provide responses based only on approved information.

##### 4. Response Testing

Create realistic test scenarios covering:

- normal supported questions;
- ambiguous questions;
- unsupported requests;
- requests requiring escalation;
- questions where information is unavailable.

## 5. Initial Prototype

Develop an initial working prototype or demonstrable assistant workflow.

### Week 5 Output

Prepare an **Initial HealthConnect AI Assistant Prototype Package** containing:

- Prompt library.
- Use-case test cases.
- Sample outputs.
- Knowledge-base usage approach.
- Safety testing.
- Escalation testing.
- Prototype/workflow.
- Limitations and next steps.

## 9. CROSS-TRACK COLLABORATION

HealthConnect is a shared multidisciplinary project.

During Week 5, interns should identify at least **one meaningful dependency or collaboration point** with another track.

Examples:

- Data Analytics → provides insights/KPIs that can inform Data Science.
- Data Science → provides modelling requirements that can inform ML Engineering.
- ML Engineering → provides implementation requirements for the eventual ML solution.
- Generative AI → uses approved HealthConnect information and may require project-management coordination.
- Project Management → coordinates dependencies and progress across all tracks.

### Required Documentation

Briefly document:

1. Which track you interacted with.
2. What information/output was exchanged.
3. Why the interaction was relevant.
4. What changed or improved as a result.

This is intended to simulate a **real multidisciplinary project environment**.

## 10. PART 3: ASSUMPTIONS, LIMITATIONS, RISKS & DEPENDENCIES

Review the assumptions, limitations, risks and dependencies identified in Week 4.

Document:

- Which remain relevant.
- Which have been resolved.
- Which have changed.
- Any new issues discovered during Week 5.
- Their potential impact on the project.
- How they should be addressed.

Do not create a separate report if this information can be incorporated into your main track-specific submission.

## 11. PART 4: WEEK 5 PROJECT SUMMARY

Prepare a concise Week 5 Project Summary covering:

1. **What you planned to accomplish in Week 5.**
2. **What you actually completed.**
3. **Key findings or development outcomes.**
4. **Major challenges encountered.**
5. **Important decisions made and why.**
6. **Any changes to your Week 4 approach.**
7. **Cross-track collaboration completed.**
8. **Remaining work.**
9. **Your proposed focus for Week 6.**

The summary should complement your main project documentation rather than repeat it.

## 12. SUBMISSION REQUIREMENTS

Your Week 5 submission must include:

### 1. Track-Specific Week 5 Output

Submit the required output for your assigned track.

### 2. Week 5 Project Summary

Submit the concise project summary described above.

### 3. Supporting Files

Include relevant supporting files such as:

- notebooks;
- datasets/processed datasets;
- reports;
- dashboards;

- code;
- project plans;
- prototypes;
- testing evidence;
- documentation.

#### 4. Google Drive

Upload the required Week 5 files to your designated project folder.

Ensure:

- correct files are uploaded;
- files are properly named;
- folders are organised;
- mentor access is available.

#### 5. GitHub

Where applicable:

- update the repository;
- commit Week 5 work;
- update README.md;
- maintain an organised project structure;
- ensure links/files are accessible.

### 13. PROFESSIONAL DEVELOPMENT REQUIREMENT

Continue documenting your HealthConnect project journey professionally.

Interns should:

- maintain an organised project folder/repository;
- document important decisions;
- maintain appropriate version control;
- update their README;
- present their work professionally;
- share meaningful project progress on LinkedIn and X where appropriate;
- tag **AnalystLab Africa** on LinkedIn and **analystlabafri**c on X where appropriate;
- use **#AnalystLabAfrica** where appropriate.

Any publicly shared work must accurately represent the intern's actual work and progress. This continues the professional-development expectations established in Week 4.

#### Assessment Criteria

| Assessment Criteria                                | Weight |
|----------------------------------------------------|--------|
| Understanding and application of Week 4 foundation | 15%    |

|                                          |      |
|------------------------------------------|------|
| Quality of track-specific implementation | 25%  |
| Technical/analytical quality             | 20%  |
| Problem-solving and decision-making      | 15%  |
| Documentation and organisation           | 10%  |
| Cross-track collaboration                | 5%   |
| Professional presentation                | 10%  |
| Total                                    | 100% |

## 15. IMPORTANT WEEK 5 REMINDER

Week 5 is the **development and implementation stage** of the HealthConnect Experience Lab.

You are **not expected to complete the entire HealthConnect solution this week**.

Your focus should be on:

**Review Week 4 → Implement → Analyse/Develop → Test Initial Outputs → Document → Prepare for Week 6**

Your Week 5 work should provide a clear and usable foundation for the next stage of the HealthConnect project.